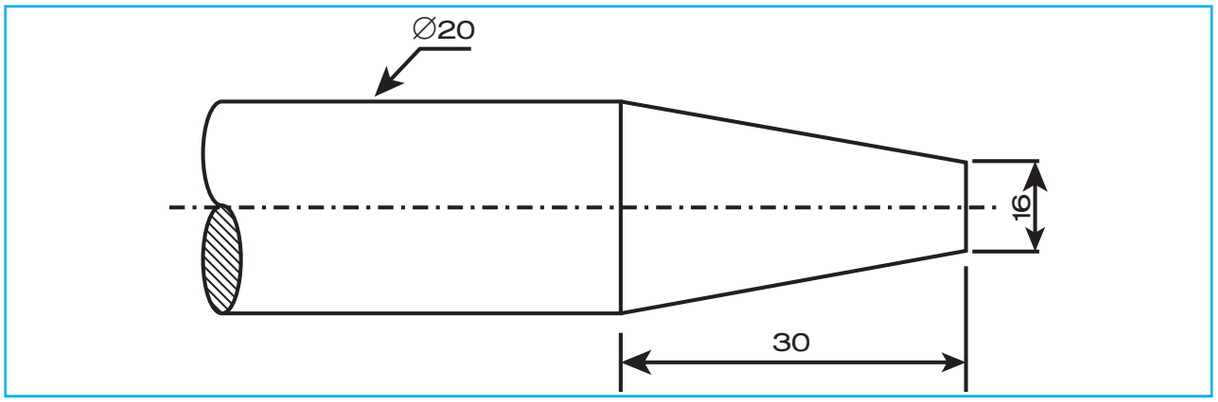
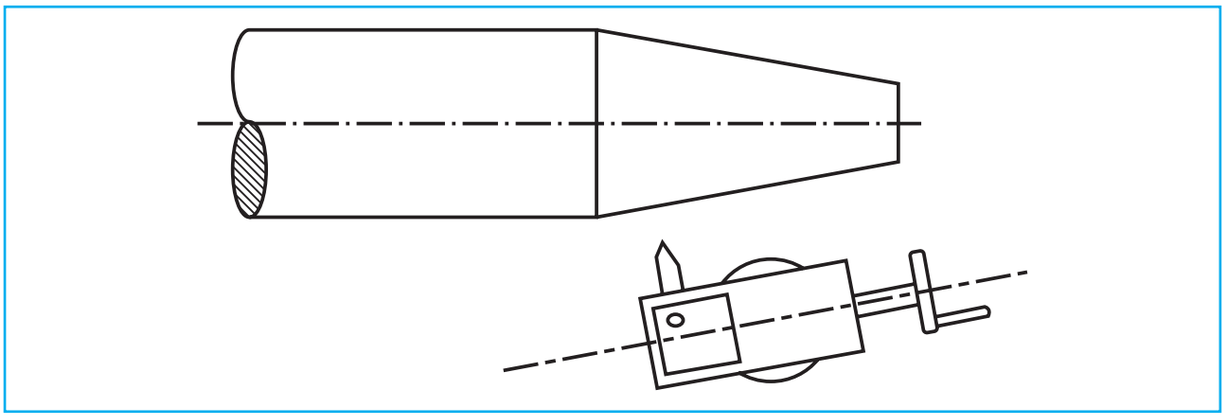
## Exercise – 5




## Operation – 1



### Marking

Marking is done on the work at the required point to indicate the length of the steps to be turned.

### Procedure

1. The angle (θ) through which the compound slide is to be swiveled is calculated by the formula

   $$\tan \theta = \frac{D-d}{2l}$$

   where
   'D' is the larger diameter
   'd' is the smaller diameter and
   'l' is the length of the taper

2. The compound slide is swiveled to the angle calculated as above.
3. The depth of cut is set by adjusting the cross-slide.
4. The plain turning tool mounted on the toolpost is given feed by the compound slide at the required angle to the lathe axis.
5. The above process is repeated until the taper required length is obtained. This is done by adjusting the cross-slide for depth of cut and compound slide for feed.
6. While doing so, feed is provided from smaller diameter to larger diameter.

### Result

The finished piece of work is sent for inspection after the machining is over.



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